OFFICE HOURS TODAY
3:30 - 4:30 PLC 530

Unbiased

↳ exp. value: $\mathbb{E}[\hat{\beta}] = \beta$
(unbiased)

↳ holding sample size fixed, what is the avg. estimate from the estimator

Consistency

↳ $\text{plim}\ \hat{\beta} = \beta$
consistent

↳ As $n \to \infty$ (sample size get "big"), does the estimator's distribution collapse to β?

Heteroskedasticity

$Var(u_i) \neq \sigma^2$ for all i (individuals)

$\Rightarrow$ some ppl have more var. than others in their disturbances

(OLS "classical assumptions" assume homosk. $\to Var(u_i) = \sigma^2$ for all ind. i)

↳ Heterosk. means
1. OLS is no longer "BLUE" (it is less efficient than WLS).
2. OLS "classic" standard errors are biased.

weights obs. inversely to σ_i^2

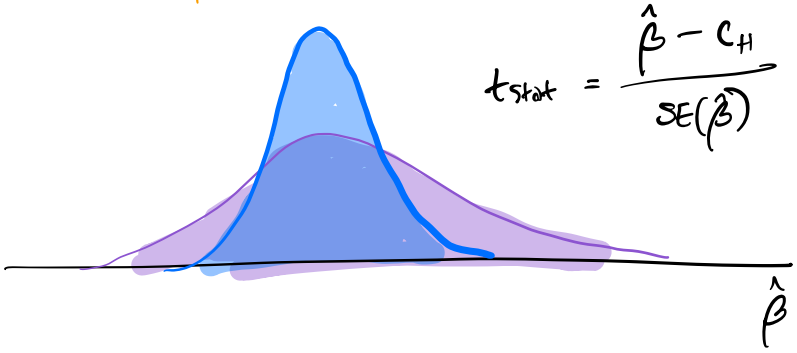
Solution: Use heterosk.-robust standard errors.

Standard dev. of an estimator's distribution.
↳ tells us about uncertainty

Ex.

$$Y_s = \beta_0 + \beta_1 X_s + u_s$$

coef	SE	t	p-val



$$t_{stat} = \frac{\hat{\beta} - c_H}{SE(\hat{\beta})}$$

$$\overset{\$}{Sales_t} = \beta_0 + \beta_1 \overset{\$}{Adv_t} + u_t$$

Clustered / correlated disturbances

"Classical" OLS assumption: $Cov(u_i, u_j) = 0$ (when $i \neq j$) $\Rightarrow$ disturbances are independent

$$\text{Test Score}_i = \beta_0 + \beta_1 \text{Hrs Studied}_i + u_i$$

= professor
temperature
Super Bowl
clicking

↳ Causes OLS to be biased for SEs

$\Rightarrow$ Use "clustered" or "cluster-robust" standard errors.

Omitted variable bias

Requires two things

1. omitted variable affects the outcome.
2. omitted correlates w/. an included regressor.

$$\text{Test Score}_i = \beta_0 + \beta_1 \text{Hrs Studied}_i + u_i$$

Super Bowl$_i$

image SB has (+) on Tests
(−) on Study

$\Rightarrow \text{plim}\ \hat{\beta}_1 = \beta_1 + (-)$

OLS is biased downward here.

OVB Formula

$$\text{plim}\ \hat{\beta}_1 = \beta_1 + \beta_2 \frac{Cov(x_1, x_2)}{Var(x_1)}$$

effect of included var. x_1

(?) (?) (+)

effect of omitted var x_2

Time Series

1. Static model: $y_t = \beta_0 + \beta_1 x_t + u_t$ (ignores cross-period effects)
2. Dynamic model w/ lagged expl. vars: $y_t = \beta_0 + \beta_1 x_t + \beta_2 x_{t-1} + \beta_3 x_{t-2} + u_t$
3. ADL: $y_t = \beta_0 + \beta_1 y_{t-1} + \beta_2 x_t + \beta_3 x_{t-1} + u_t$

Autoregressive distributed lag models

OLS is biased but consistent for ADL models (w/ lagged outcomes).

Autocorrelation for the disturbance: AR(1) $u_t = \rho u_{t-1} + \varepsilon_t$

↳ Has very similar impacts for OLS as heterosk.

- Biases SE estimates for 1 and 2.
- Causes OLS to be inefficient

New: OLS is inconsistent for 3 when u_t is autocorrelated